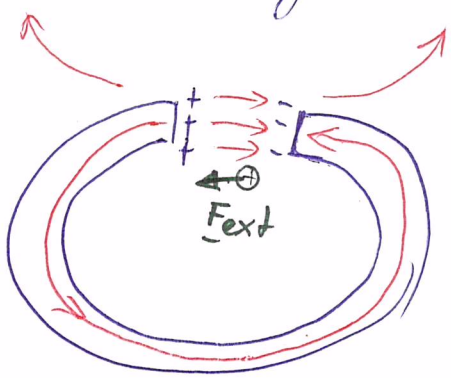
7. DC circuits

To maintain a stationary flow of charge (DC) we need to provide charge carriers and energy continuously:



$$\oint \underline{E}\, d\underline{r} = 0$$

You have to move charge carriers against the electric force by some external force!

$$\underline{F}_{ext} = -\underline{E}\, q$$

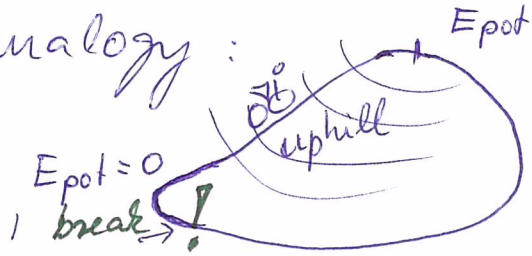
$$\parallel$$

$$\underline{E}^* \cdot q$$

What can be this external force

- Chemical force in batteries (difference in electro-negativity)
- Mechanical -"- in van der Graaf generator (moves the belt)
- Magnetic force in a dynamo or alternator (electromagnetic induction)

Bicikle analogy:

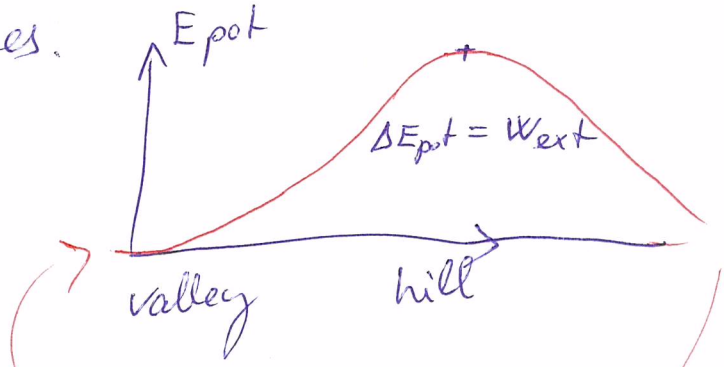


downhill: easy, gravity accelerates you, work made by the field (gravity) is positive, your pot. energy decreases. You arrive to the valley with plenty of kinetic energy, that brings you back to the top of the hill. Is that reality?

No! because of drag forces. a large part of your.

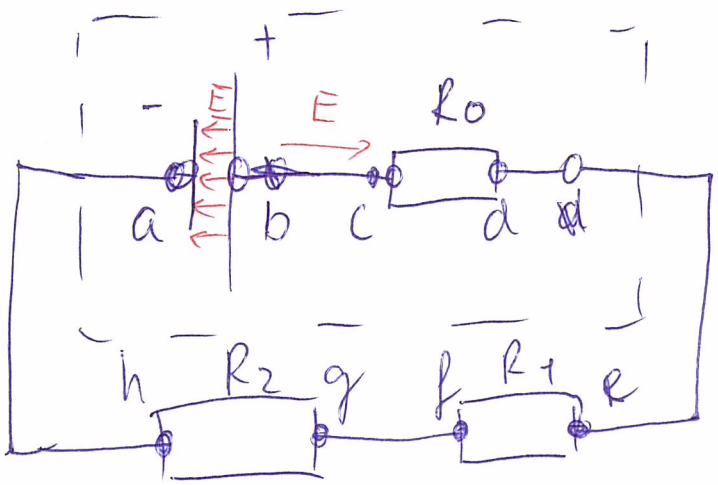
Drag forces will dissipate ~~you~~ kinetic energy, so you arrive to the bottom at a low speed.

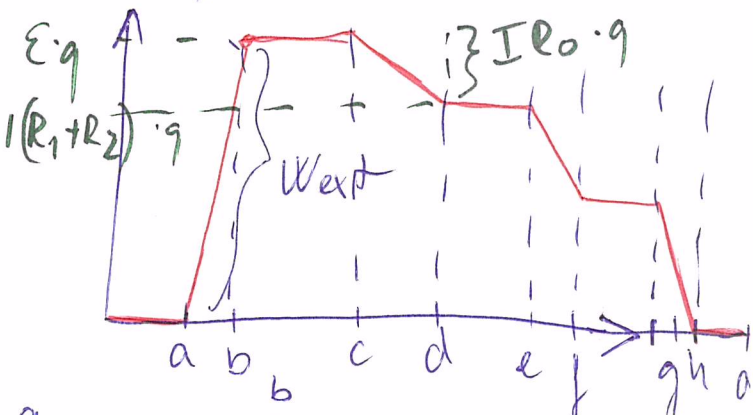
How to go up? You need external (not gravitational) force: the force of your muscles! With your external force you make positive work ($\underline{F}_{ext} \parallel d\underline{s}$), your pot. energy increases.



~~ΔU~~ (ΔE_{pot}) $= \Delta U \cdot q$

For a DC current:





$$E_{pot}^{a}(b) = -\int_a^b \underline{E}\, d\underline{r} \cdot q > 0 \quad \text{positive!}$$

$\underline{E} \uparrow\downarrow d\underline{r}$

$$E_{pot}^{a}(b) = W_{ext} = \int_a^b \underline{F}_{ext} \cdot d\underline{r} = \int_a^b \underline{E}^{*}_{ext} \cdot d\underline{r} \cdot q$$

$\underline{F}_{ext} \uparrow\uparrow$

$\underline{E}^{*}_{ext}$ virtual field of ext force

Electroactive force (EMF)

$$\mathcal{E} = \int_a^b \underline{E}_{ext}\, d\underline{r} = \text{virtual ~~potential~~ diff. of ext force}$$

$\Delta U_{battery} = \mathcal{E} - IR_0$: EMF - voltage drop on internal resistance !

Ohm's law $I = \dfrac{\mathcal{E}}{R_0 + (R_1 + R_2)}$

$\mathcal{E} = IR_0 + I(R_1 + R_2 + \dots)$ / · I

$\mathcal{E}I = I^2 R_0 + I^2 \cdot (R_1 + R_2)$

$I^2 R_0$: loss in battery; $I^2 \cdot (R_1 + R_2)$: effective work done

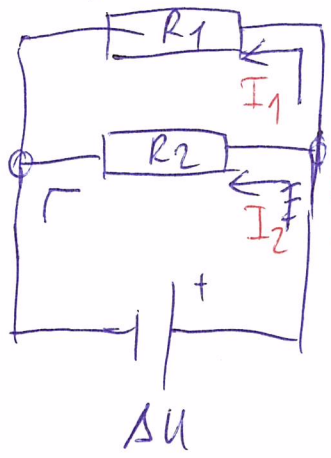
(by convention $\mathcal{E} > 0$)

Resistors in series

$$\mathcal{E} = I(R_1 + R_2 + R_3 + \dots) = I(R_{eq})$$

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

in parallel:



$\uparrow I_1 + I_2$ (Kirchhoff I)

$I_1 R_1 = \Delta U$

$I_2 R_2 = \Delta U \dots$

$$I_1 + I_2 = I = \frac{\Delta U}{R_1} + \frac{\Delta U}{R_2} + \dots$$

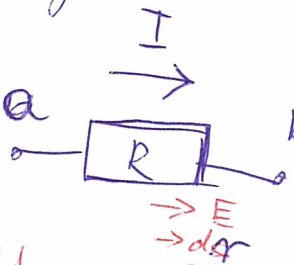
$$R_{eq} = \frac{\Delta U}{I} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots}$$

} household: always connected in ||

Kirchhoff's 2nd rule:

$$\sum_{\text{closed loop}} \Delta U = 0 \iff \oint \underline{E}\, d\underline{r} = 0$$

Sign conventions (suppose I and $\mathcal{E}$ is positive)

① 

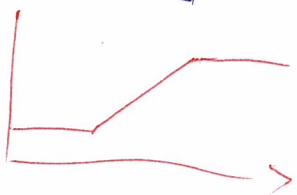
$$\Delta U^{a}(b) = -\int_{a}^{b} \underline{E}\, d\underline{r} = -IR < 0$$

$$\left(= -\int_{a}^{b} \frac{\underline{j}}{\sigma} d\underline{r} = -\frac{\underline{j}\cdot l}{\sigma} = -\frac{I\,l}{A\,\sigma} = -IR \right.$$

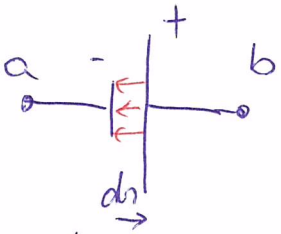


② 

$$\Delta U^{a}(b) = IR > 0$$

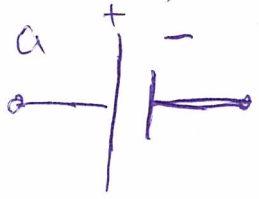


In the direction of current flow voltage drops!

③ 

$$\Delta U^{a}(b) = -\int \underline{E}\, d\underline{r} = \mathcal{E} > 0$$

$\underline{E} \parallel d\underline{r}$

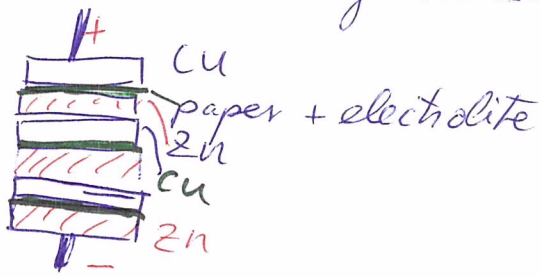
④ 

$$\Delta U^{a}(b) = -\int \underline{E}\, d\underline{r} = -\mathcal{E} < 0$$

In an ideal battery (internal resistance = 0)
voltage drops from + to - electrode,
voltage ~~differs~~ magnitude = $\mathcal{E}$ EMF

Story : who has discovered the electric battery?

The first electrochemical battery (Voltaic pile)
was made by Alessandro Volta in 1800



1792

Argument :
electricity comes from the animal or from the different metals ??

Luigi Galvani 1791

- animal electricity
- touching the frog's legs by ~~metals~~ metallic electrodes → muscles move

Hun : galvánelem (galvanic cell)
galvanization ?

Michael Faraday : understood the operation, explained by chemical reactions

Electrical safety ?